

Forecasting and Optimization for Industrial Energy Management

Project Report

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October 2026

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First narrative draft. All reported forecasting and economic results use frozen 2024-to-2025 experiments. The deployment and monitoring section describes the implemented training layer and the intended operating model; it does not claim that a production platform has already been deployed.

Executive summary

This project develops a small but complete decision system for an industrial electricity consumer with flexible production load, on-site photovoltaic generation and battery storage. The system must decide how to distribute 200 kWh of daily production over a 06:00–22:00 operating window, how to use a 44.16 kWh battery, how much electricity to nominate in the German day-ahead market, and how to revise the physical schedule as new information becomes available during the day.

The technical core combines point and probabilistic forecasting, mixed-integer optimization, receding-horizon model-predictive control and scenario-based stochastic optimization. The forecasting experiments include gradient-boosted trees, persistence and autoregressive baselines, direct residual correction, and MIMO models based on multilayer perceptrons and LSTMs. Uncertainty is represented through joint residual bootstrap, multi-quantile regression, empirical copulas and conformal calibration. The decision experiments compare deterministic mixed-integer programming, sample average approximation and mean-CVaR optimization. This breadth is deliberate: the project tests an integrated decision workflow and the practical value of advanced methods, rather than a single isolated model.

All model and policy choices are developed on 2024 data and then frozen. The final evaluation uses 261 complete delivery days from January through September 2025. Forecast metrics are reported as diagnostics, while the main criterion is realised economic cost. Every policy is executed and settled against the same factual PV generation, day-ahead prices, intraday-price proxy, grid tariffs and battery-cost assumptions.

Two reference policies frame the economic comparison. The **Rule-based strategy** is a performance floor: a common-sense schedule derived from observed market dynamics, placing flexible load in historically cheaper hours and battery discharge in historically expensive hours. It requires no machine learning or

mathematical optimization and could be implemented in a spreadsheet. The **Day-ahead Oracle** is the performance ceiling and cost lower bound. It allocates load, PV and battery capacity using factual PV generation and factual day-ahead prices that would not be known when a real nomination is submitted. The Oracle is still prevented from speculative intraday trading, so its advantage comes only from perfect day-ahead information.

The resulting costs and the share of the Rule-to-Oracle opportunity captured by each evaluated strategy are presented below.

Strategy	Jan-Sep 2025 cost	Saving vs Rule	Opportunity captured
Rule-based	EUR 7,992.53	–	0.0%
Point day-ahead	EUR 7,547.38	EUR 445.15	79.4%
Stochastic day-ahead, raw residual SAA	EUR 7,535.12	EUR 457.41	81.6%
Point day-ahead + MPC	EUR 7,484.07	EUR 508.46	90.7%
Day-ahead Oracle	EUR 7,431.97	EUR 560.56	100.0%

Economic value of forecasting and optimization

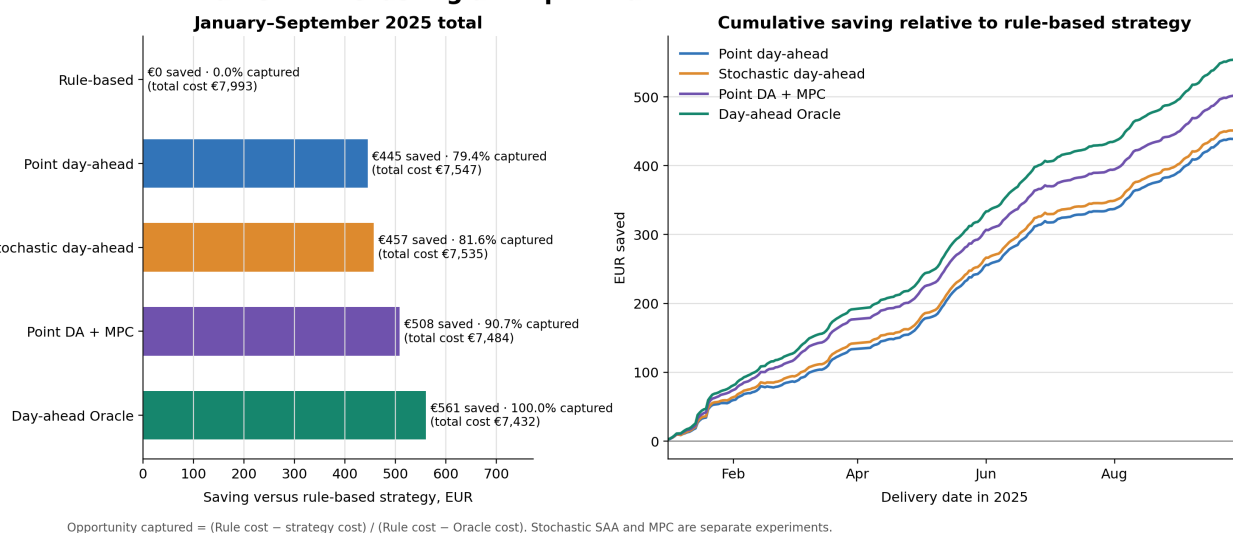


Figure 1. Economic value relative to the common Rule-based policy. The stochastic day-ahead and deterministic MPC results are separate experiments; their gains must not be added.

The result should be read as evidence about the architecture and the relative value of its components, rather than as a commercial savings claim. Access to data was limited to public measurements, archived weather forecasts, reanalysis and public market indices. The project has no on-site weather station, plant telemetry, executable intraday quotes or order-book data. The PV plant, battery and industrial demand form a synthetic scenario around a real measured PV shape, and the economic replay covers nine months. Developing state-of-the-art standalone forecasting models was therefore not the primary goal: the models were kept deliberately compact so that the full path from historical data to forecast, constrained decision and realised settlement could be evaluated consistently.

Motivation

The project began as a personal applied-research exercise: to combine data science, forecasting, optimization and MLOps around one energy-management problem. A parallel goal was to understand the structure of the German electricity market and to represent its relevant operational and settlement rules explicitly in the decision model.

The objective was not to reproduce a commercial energy-management platform or to imply that a public-data simulation can determine an investment case. The objective was to formulate the information boundary carefully, build the decision chain end to end, compare strategies of increasing sophistication, and

identify where added complexity creates measurable value.

1. Business setting and decision chronology

The model represents an industrial prosumer in the DE-LU market area. The production process requires 200 kWh during local hours 06:00–21:00. Each hour must consume between 5 and 20 kWh, but the exact profile is flexible. A 10 kWp PV proxy and two reference battery units can offset grid purchases or export energy.

The decision process has two layers.

Day-ahead decision

Before delivery, the system receives archived weather forecasts, historical prices, the initial battery state and the production requirement. It predicts PV generation and the next day's day-ahead price curve, then jointly selects:

- the hourly production-load profile;
- battery charge, discharge and state of charge;
- PV curtailment;
- the signed day-ahead market position.

The resulting position is fixed after the auction.

Intraday MPC

Immediately before 06:00 and then once per hour, the controller observes the executed load, battery state and completed PV and intraday-price observations. It updates the PV and price forecasts, solves the remaining-horizon problem, executes only the next action, and repeats. Past actions and the day-ahead position cannot be changed.

The realised difference between physical grid exchange and the fixed day-ahead position is settled at an hourly intraday continuous-price proxy. Grid charges are applied to physical import, rather than to the financial market deviation.

2. Data and reference scenario

The project combines measured, public and synthetic inputs.

Component	Source and role	Main limitation
PV output	KIT MPVBench profile 1a, 15-minute measurements, aggregated hourly	Exact panel coordinates and timestamp convention are not published
Weather fact	ERA5 reanalysis near Pforzheim, approximately 48.89 N, 8.70 E	Grid-cell reanalysis, not an on-site sensor
Weather forecasts	Archived DWD ICON forecasts; ten German locations for the price model	Fixed historical forecast products do not reproduce a live vendor feed exactly
Day-ahead prices	Public DE-LU auction history	Hourly products end after September 2025 in the working data contract
Intraday prices	Energy-Charts Intraday Continuous Average Price	Delivery-period average, not an executable quote or order book
Production load	Synthetic flexible demand	No real plant process or ramp constraints
Battery and tariff	Reference equipment and Pforzheim 2025 variable tariff	Engineering and contract assumptions, not site telemetry or an offered supply contract

The common UTC intersection of actual weather, archived weather forecasts and day-ahead prices covers 17,543 hourly timestamps in 2024–2025. MPVBench contains 70,176 fifteen-minute records across five profiles. Profile 1a is used because it has the cleanest coverage; three dates with simultaneous zero production across several profiles are treated as data outages.

The PV series provides a real production *shape*. It is scaled by a factor of 20 from the roughly 0.5 kW source profile to a 10 kWp modelling asset. The working timezone assumption is Europe/Berlin, supported by the observed solar-day alignment but not confirmed by source metadata.

The fixed physical scenario is:

Parameter	Value
Daily production energy	200 kWh
Operating window	06:00–22:00, 16 delivery hours
Hourly production bounds	5–20 kWh
PV proxy	10 kWp
Battery usable capacity and initial SoC	44.16 kWh
Maximum charge and discharge	5 kWh per hour
Charge/discharge efficiency	98% / 98%
Degradation cost	EUR 0.05 per discharged kWh
Daytime variable import tariff	EUR 0.10181 per kWh
Nighttime variable import tariff	EUR 0.07441 per kWh

Fixed daily network charges are excluded from dispatch and pairwise strategy comparisons because they are identical for every policy. They can be added to an absolute customer bill without changing any decision or relative saving.

3. Mathematical decision model

3.1 Notation

Let the operating hours be

$$\mathcal{H} = \{6, 7, \dots, 21\}.$$

For hour h , let L_h be production consumption, G_h available PV, U_h curtailed PV, C_h battery charge, D_h battery discharge and E_h state of charge. The signed physical grid exchange is

$$g_h = L_h + C_h - (G_h - U_h) - D_h.$$

Positive g_h is import and negative g_h is export. Its linear split is

$$I_h \geq g_h, \quad I_h \geq 0,$$

$$X_h \geq -g_h, \quad X_h \geq 0.$$

Here q_h^{DA} is the signed day-ahead nomination and

$$\Delta_h = g_h - q_h^{DA}$$

is the realised intraday deviation.

3.2 Flexible production load

The process must complete its daily energy requirement while respecting hourly bounds:

$$\sum_{h \in \mathcal{H}} L_h = 200,$$

$$5 \leq L_h \leq 20 \quad \forall h \in \mathcal{H}.$$

This is intentionally an energy-allocation model. Start-up costs, product deadlines, ramping and machine-level constraints are outside the V1 scope.

3.3 PV and curtailment

PV may be consumed, stored, exported or curtailed:

$$0 \leq U_h \leq G_h.$$

Curtailment prevents the optimizer from treating export during sufficiently negative prices as mandatory. In execution, the optimized curtailment fraction is applied to factual PV so that forecast error cannot create infeasible negative generation.

3.4 Battery dynamics

For charge efficiency η_c and discharge efficiency η_d ,

$$E_{h+1} = E_h + \eta_c C_h - \frac{D_h}{\eta_d},$$

$$0 \leq E_h \leq E^{\max}, \quad E^{\max} = 44.16.$$

Charge and discharge are capped at 5 kWh per hour. A binary operating mode prevents simultaneous charging and discharging:

$$0 \leq C_h \leq 5z_h,$$

$$0 \leq D_h \leq 5(1 - z_h), \quad z_h \in \{0, 1\}.$$

The model can charge during the working day when wholesale price, grid tariff, PV opportunity cost and terminal battery value make it economic. This includes negative-price periods and low-price PV surplus.

Overnight charging is not represented as a separate hourly optimization problem. Instead, each operating day starts with a full battery, and any closing SoC shortfall is converted into a terminal recharge liability using a causal proxy for the expected overnight charging price:

$$C_T = \frac{E^{\max} - E_T}{\eta_c} \left(\frac{\max(\bar{P}^{night}, 0)}{1000} + t^{night} \right).$$

The causal night-price benchmark $\bar{P}^{night}$ is the rolling mean of the latest 14 completed nights. Each night covers price labels 22:00, 23:00 and 00:00–06:00. Negative nightly means are floored at zero before aggregation.

3.5 Settlement ledger

Let p_h^{DA} and p_h^{ID} be realised prices in EUR/MWh, t_h^{imp} the variable physical-import tariff, t^{exp} the export charge and c^{deg} battery degradation cost. Realised daily cost is

$$C_{day} = \sum_h \frac{P_h^{DA}}{1000} q_h^{DA} + \sum_h \frac{P_h^{ID}}{1000} \Delta_h \\ + \sum_h t_h^{imp} I_h + \sum_h t_h^{exp} X_h + c^{deg} \sum_h D_h + C_T.$$

The ledger separates the financial market position from physical grid use. This avoids charging network fees on a purely financial deviation and supports exact reconciliation of every strategy.

3.6 Deterministic day-ahead optimization

The deterministic optimizer replaces future PV and prices by point forecasts $\hat{G}_h$ and $\hat{P}_h^{DA}$, solves the physical schedule and sets

$$q_h^{DA} = L_h + C_h - (\hat{G}_h - U_h) - D_h.$$

The decision variables are the hourly load, charge, discharge, curtailment, SoC, battery mode and physical import/export variables. The optimizer minimizes forecasted day-ahead energy cost, physical grid charges, battery degradation and terminal recharge liability:

$$\min_{L,C,D,U,E,z,I,X} \left[\sum_{h \in \mathcal{H}} \left(\frac{\hat{P}_h^{DA}}{1000} q_h^{DA} + t_h^{imp} I_h + t_h^{exp} X_h + c^{deg} D_h \right) + C_T \right].$$

The binary battery-mode variable z_h makes this a mixed-integer linear program rather than a continuous linear program. It is required to enforce the discrete choice between charging and discharging in each hour.

The schedule is therefore balanced under the information available before the auction. Forecast error creates a deviation only when the day is executed.

3.7 Shrinking-horizon MPC

At decision time τ , all actions before τ , current SoC and the completed portion of the 200 kWh requirement are fixed. The controller solves the same model over

$$\mathcal{H}_\tau = \{\tau, \tau + 1, \dots, 21\},$$

subject to the remaining-load constraint

$$\sum_{h \in \mathcal{H}_\tau} L_h = 200 - \sum_{h < \tau} L_h^{actual}.$$

The day-ahead position q_h^{DA} remains fixed. Load, charge, discharge, curtailment and SoC are reoptimized. Only the first action is executed before the horizon shrinks and the problem is solved again.

At time τ , the MPC decision variables are the same physical variables as in the day-ahead problem, restricted to the remaining horizon. The incurred day-ahead purchase is already fixed and is therefore a constant in this optimization. MPC minimizes the forecasted cost of the remaining intraday deviations, physical grid use, battery degradation and terminal recharge:

$$\min_{L,C,D,U,E,z,I,X} \left[\sum_{h \in \mathcal{H}_\tau} \left(\frac{\hat{P}_{h|\tau}^{ID}}{1000} (g_{h|\tau} - q_h^{DA}) + t_h^{imp} I_h + t_h^{exp} X_h + c^{deg} D_h \right) + C_T \right].$$

The next-hour intraday-price forecast uses the learned spread correction; more distant hours retain the known day-ahead curve. The forecast grid exchange $g_{h|\tau}$ uses the latest PV forecast and current factual battery state.

3.8 Stochastic day-ahead optimization

The stochastic extension generates $S = 500$ joint daily scenarios for PV, day-ahead price and the intraday-minus-day-ahead spread. Two scenario-generation approaches were implemented and compared.

The intraday spread is needed even though the decision is made day-ahead. Errors in the PV forecast cause the realised physical exchange to differ from the submitted day-ahead position, and that deviation is settled at the future intraday price,

$$P_{s,h}^{ID} = P_{s,h}^{DA} + S_{s,h}.$$

The spread scenarios therefore value forecast-induced imbalance risk; they do not permit the optimizer to take a speculative intraday position. If the model instead assumed $P_h^{ID} = P_h^{DA}$ in every scenario, a separate spread forecast would not be required.

1. **Joint residual bootstrap.** Rolling-origin 2024 forecast errors are stored as complete daily trajectories. One sampled historical day supplies the PV, day-ahead-price and spread residual paths together, preserving cross-hour and cross-target dependence. Raw, centered and seasonally weighted variants were evaluated.
2. **Multi-quantile regression with an empirical copula.** Separate conditional quantile models estimate the marginal predictive distributions. Historical rank vectors form an empirical copula that reconnects the marginals into coherent multivariate daily scenarios. Conformal calibration was evaluated as an additional coverage correction.

Both generators feed the same stochastic optimizer, which separates the value of scenario construction from the value of the optimization formulation.

The physical decisions are shared across scenarios. To prevent deliberate day-ahead/intraday speculation, the nomination remains tied to the frozen point PV forecast. Scenario uncertainty changes the settlement deviation, not the commercial interpretation of the nomination.

Sample-average approximation solves

$$\min_x \frac{1}{S} \sum_{s=1}^S C_s(x).$$

The risk-averse extension adds empirical CVaR:

$$\min_{x, \zeta, \xi_s} \left[\frac{1}{S} \sum_s C_s(x) + \lambda \left(\zeta + \frac{1}{(1-\alpha)S} \sum_s \xi_s \right) \right],$$

subject to

$$\xi_s \geq C_s(x) - \zeta, \quad \xi_s \geq 0,$$

with $\alpha = 0.95$. Stochastic MPC is deliberately excluded: it would require hundreds of scenario optimizations at every hourly control step, while the deterministic MPC already leaves a small residual opportunity.

3.9 Rule-based and Oracle boundaries

The Rule-based policy uses an average historical price shape to place high load in cheap hours and battery discharge in expensive hours. Its day-ahead purchase covers the production profile. Factual PV can refill battery headroom, serve load or be exported; residual PV is curtailed under negative prices.

The Oracle knows factual PV and factual day-ahead prices before nomination. It does not know or trade against future intraday prices. Its nomination equals its physical schedule,

$$q_h^{DA} = g_h,$$

so $\Delta_h = 0$. This defines a useful lower cost bound without giving the benchmark an artificial speculative business model.

4. Forecasting and uncertainty models

The point models are deliberately compact. The project aims to test an entire decision system, rather than to win a forecasting benchmark through extensive model search.

Decision input	Final point model	Frozen 2025 result
Day-ahead PV	CatBoost with calendar, solar geometry and archived weather	MAE 26.17 W, RMSE 51.35 W; nMAE 5.32% of 2024 observed peak
Intraday PV	Direct CatBoost residual correction	MAE 16.41 W vs 17.44 W frozen-head baseline on MPC pairs
Day-ahead price	CatBoost with price history, calendar and spatial German weather	MAE 17.43 EUR/MWh, RMSE 27.31 EUR/MWh, daily Spearman 0.912
Next-hour intraday price	Regularized CatBoost on intraday-minus-day-ahead spread, shrunk 70%	MAE 12.67 vs 12.79 EUR/MWh for the known day-ahead curve

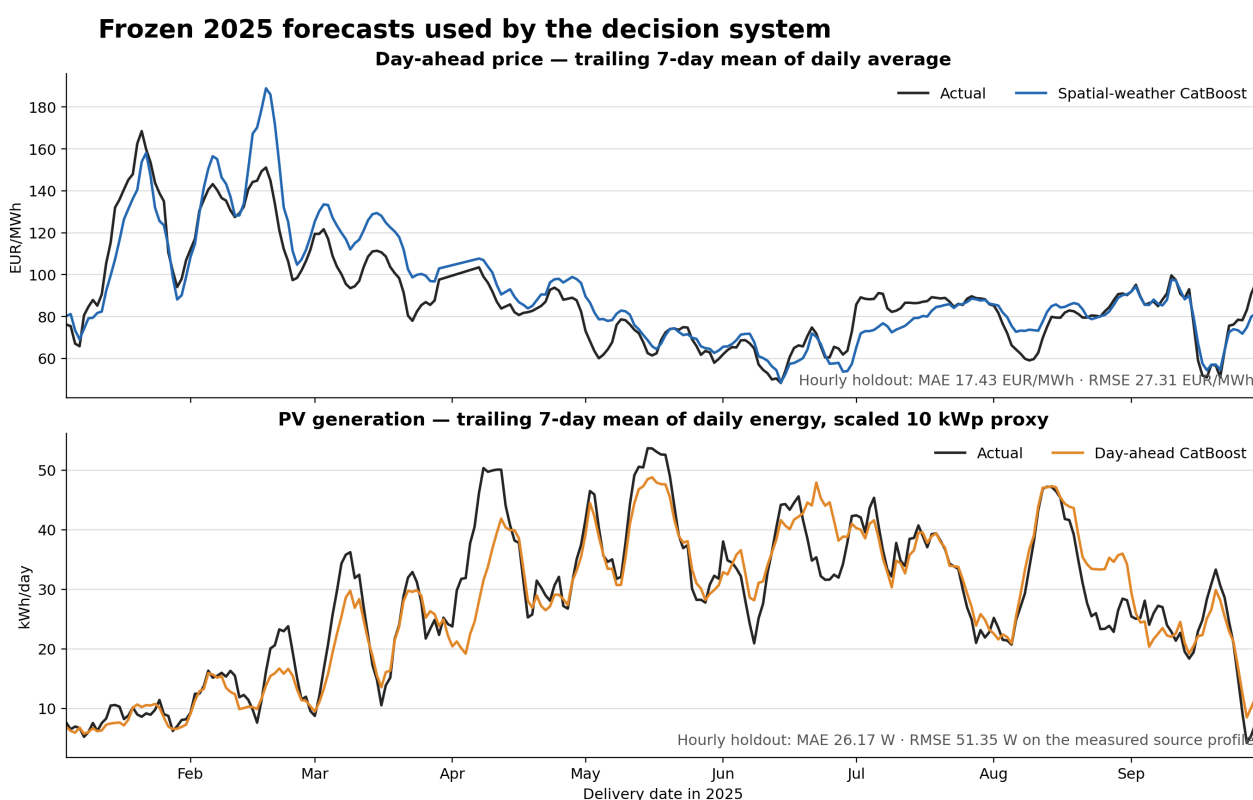


Figure 2. Frozen forecasts against 2025 facts. The lines are trailing seven-day summaries for readability; reported metrics are calculated on hourly observations.

4.1 PV model comparison

The intraday PV experiment compares a frozen day-ahead forecast, latest-error persistence, direct CatBoost residual correction, a MIMO multilayer perceptron and a MIMO LSTM. Every model is evaluated on the same 17,034 (decision time, future target hour) pairs.

Intraday PV method	MAE, W	RMSE, W	MAE change vs frozen day-ahead
Frozen day-ahead forecast	17.44	33.15	–

Intraday PV method	MAE, W	RMSE, W	MAE change vs frozen day-ahead
Latest-residual persistence	44.65	68.59	156.0% worse
Direct residual CatBoost	16.41	31.70	5.9% better
MIMO MLP	17.28	33.11	0.9% better
MIMO LSTM	17.08	33.73	2.1% better

The difference between CatBoost and the neural trajectory models is modest, but CatBoost is both the most accurate and the simplest model to train, serve and explain. More data-intensive sequence architectures, including temporal transformers such as TFT, Informer or PatchTST, were considered but not pursued. With only one year of single-site training history, a reliable comparison would be difficult and added model capacity would be more likely to increase variance than to improve the decision system.

4.2 Price model comparison

The day-ahead price experiment moves from a simple previous-day persistence baseline to price-history CatBoost, then adds local and spatial weather forecasts.

Day-ahead price method	MAE, EUR/MWh	RMSE, EUR/MWh	Mean daily Spearman
Previous-day same-hour persistence	25.57	40.12	0.796
Price history + calendar	20.36	30.95	0.876
Price history + local weather	19.33	30.06	0.878
Price history + spatial German weather	17.43	27.31	0.912

For intraday prices, the already-cleared day-ahead curve remains the strongest baseline. The tested alternatives include last-price persistence, direct CatBoost prediction of the intraday-minus-day-ahead spread, richer rolling statistics, a local AR(1) feature, and updated local and spatial weather. The selected hybrid applies a shrunk CatBoost spread correction only to the next delivery hour and retains the day-ahead curve beyond it. The full-horizon MAE improves only from approximately 12.8 to 12.6 EUR/MWh; applying the correction to every remaining hour is worse. The final design therefore keeps the small, stable correction rather than the most elaborate feature set.

4.3 Probabilistic forecasting and scenario generation

The project evaluates conditional quantile regression and residual-based scenario generation, rather than treating uncertainty as a fixed Gaussian error around the point forecast.

Approach	Construction	Observed role in the project
MultiQuantile CatBoost	Jointly estimates nine conditional quantiles from P05 to P95 for every target and hour	Flexible marginal distributions, but raw price and spread intervals are underdispersed

Approach	Construction	Observed role in the project
Empirical copula	Samples historical 48-dimensional rank vectors covering PV generation, day-ahead price and the intraday-minus-day-ahead price spread over sixteen hours	Restores cross-hour and cross-target dependence to separately estimated marginals
Conformalized quantile regression	Expands quantile intervals using rolling-origin 2024 nonconformity scores	Repairs coverage, but produces conservative intervals and no superior economic policy
Joint residual bootstrap	Samples complete rolling-origin daily residual trajectories for PV, day-ahead price and spread	Simpler dependence model and the best realised stochastic economic result
Seasonal residual bootstrap	Reweights residual days by circular calendar distance and rescales PV errors locally	Useful conditional sensitivity check, without a material economic gain

For the raw quantile model, P10–P90 coverage is 72.7% for PV, 41.2% for day-ahead price and 55.0% for intraday spread, compared with a nominal 80%. Conformal calibration raises those values to 90.9%, 87.1% and 87.7%, respectively. It solves undercoverage but overshoots the target and widens the market-price intervals substantially. The empirical-copula workflow is therefore technically valid and demonstrates conditional probabilistic forecasting, but its scenarios do not outperform the simpler residual bootstrap in the final economic replay.

The seasonal bootstrap was introduced to avoid drawing a winter residual day for a summer forecast with the same probability as a nearby calendar day. For target date d , historical residual day j receives weight

$$w_j(d) = (1 - \rho) \frac{\exp[-\Delta_{\text{circ}}(d, j)^2 / (2b^2)]}{\sum_k \exp[-\Delta_{\text{circ}}(d, k)^2 / (2b^2)]} + \rho \frac{1}{N},$$

where Δ_{circ} is circular day-of-year distance, $b = 25$ days and $\rho = 0.15$. The 15% uniform mixture prevents rare residual regimes from disappearing completely. One selected day still supplies the complete paired PV, day-ahead-price and spread trajectory. PV residuals are standardized by local hour-specific scale before being mapped to the target day's forecast scale.

This conditioning reduces the mean effective residual library from 210 to 64.8 days. It brings day-ahead-price P10–P90 coverage to 80.4%, but does not improve PV CRPS and worsens spread CRPS. Seasonal SAA saves EUR 8.81 relative to deterministic day-ahead, compared with EUR 12.26 for raw uniform residual SAA. On the May–September period with adequate seasonal support, seasonal and centered uniform sampling save EUR 3.76 and EUR 3.74 respectively. The seasonal kernel is therefore retained as a robustness check, not as the selected scenario generator.

5. Evaluation design

The evaluation follows a strict temporal contract:

- all model development and fitting use 2024 data;
- fixed configurations are refitted on all eligible 2024 observations;
- January–September 2025 is read once as a future holdout;
- no 2025 target is used in feature scaling, model selection or scenario construction;
- rolling-origin predictions create the 2024 residual library used by the stochastic generator;
- all economic policies use the same 261 complete delivery days and the same factual settlement ledger.

The price evaluation ends on 30 September because the working day-ahead source changes from hourly to 15-minute products on 1 October 2025. Quarter-hour prices are not averaged into a synthetic hourly contract.

Forecast metrics are supporting diagnostics. The primary selection criterion is realised out-of-sample economic cost. Economic differences for stochastic policies are accompanied by moving-block bootstrap

intervals, preserving some serial dependence between neighbouring delivery days.

Repeated experiments against the same 2025 period create a real risk of researcher overfitting. Results are therefore interpreted as project evidence, not as a final unbiased estimate for a commercial deployment.

6. Economic results and interpretation

6.1 Deterministic decision stack

Point day-ahead optimization saves EUR 445.15 relative to Rule-based and leaves EUR 115.41 of the Rule-to-Oracle opportunity unresolved. MPC adds EUR 63.31, closing 54.9% of that remaining gap. The combined deterministic system ends only EUR 52.10 above Oracle.

This is a stronger result than the forecast metrics alone suggest. The direct PV residual correction improves trajectory MAE by only 5.9%, and the next-hour intraday-price correction is also small. MPC still creates material value because it repeatedly reallocates the remaining production requirement and battery energy after forecast errors have become observable.

6.2 Stochastic day-ahead optimization

Raw joint-residual SAA costs EUR 7,535.12 and saves EUR 12.26 relative to the point day-ahead policy. Its 95% moving-block bootstrap interval is [EUR 6.12, EUR 19.47], so the mean-cost improvement is supported within this replay. It closes 10.6% of the remaining Point-to-Oracle gap.

The gain is useful but modest. Centering residuals reduces the saving to EUR 10.71. A seasonal conditional bootstrap saves EUR 8.81 over the full period and is nearly indistinguishable from centered uniform sampling on the seasonally well-supported May–September subset.

Adding CVaR penalties does not establish a reliable reduction in realised CVaR95. Large weights make the schedule more conservative and increase total cost. The selected stochastic strategy is therefore risk-neutral SAA rather than a mean-CVaR policy.

6.3 Physical audit

Every final strategy satisfies the same physical checks:

- exactly 200 kWh of production is completed on every retained day;
- SoC stays inside 0–44.16 kWh;
- charge and discharge never overlap;
- curtailment remains between zero and factual PV;
- the Oracle has zero intraday deviation;
- hourly energy balance and daily cost components reconcile numerically.

Battery use changes materially with policy. Rule-based performs 49.4 equivalent full cycles over the replay, point day-ahead 142.4, deterministic MPC 136.8 and Oracle 144.2. Degradation and overnight restoration are included, so the optimizer does not treat battery energy as free.

7. Software and operating model

The implementation is a modular Python package rather than a collection of model-specific services. Data contracts, feature builders, training applications, scenario generators, optimizers and backtests share one codebase and one set of units and temporal conventions.

Implemented components include:

- reproducible historical-data and feature-building commands;
- packaged training applications for day-ahead PV, MPC PV and intraday price, plus a reproducible day-ahead-price experiment;
- seasonal week and chronological split utilities;
- MLflow logging and optional Model Registry version creation;
- deterministic, MPC, SAA and CVaR decision cores;
- common replay and settlement ledgers;
- saved configurations, predictions, feature importance and audit artifacts;
- automated tests for physical constraints, temporal contracts and scenario generation.

The intended deployment remains a modular monolith: one application image, four independently versioned model artifacts, scheduled batch decisions and a small API boundary. ClearML can orchestrate ingestion, feature, training and backtest tasks; MLflow remains the source of truth for experiment tracking and model versions.

Serving, Docker packaging, ClearML pipelines, dashboards, alerting and automatic champion/fallback switching are still to be implemented. They are the appropriate next phase because the modelling and decision contracts are now stable enough to monitor.

8. Limitations

The most important limitations are structural:

1. **Synthetic industrial system.** Load flexibility, PV scale and battery capacity are scenario assumptions rather than one site's telemetry.
2. **Location uncertainty.** Weather uses an approximate Pforzheim coordinate, and the PV timestamp convention remains provisional.
3. **Market proxy.** Intraday continuous average is a settlement proxy, not an executable quote available at the control instant.
4. **Simplified production process.** The model has hourly energy bounds but no machine states, ramps, batches, labour constraints or product deadlines.
5. **Limited history.** One training year restricts neural models, tail estimation and seasonal residual sampling.
6. **Incomplete customer bill.** Variable grid charges are represented, while supplier risk premiums, balancing responsibility, demand charges and contract-specific fixed fees are outside the optimized ledger.
7. **Single repeatedly inspected holdout.** January-September 2025 is temporal out-of-sample data, but it has been used for several project-level comparisons.

These limitations do not invalidate the architecture experiment. They limit the interpretation of the absolute euro values and prevent the result from being treated as an investment recommendation.

9. Conclusion

The project demonstrates an end-to-end forecasting and optimization workflow in which models are evaluated through the decisions they support. The strongest result is the deterministic stack: point day-ahead optimization captures most of the available value, and MPC recovers more than half of the remaining gap.

Probabilistic forecasting and stochastic optimization add a smaller but measurable mean-cost improvement. Their main contribution is methodological: they expose uncertainty explicitly, preserve joint temporal dependence and allow the cost of risk aversion to be tested. The experiments also show when that sophistication is not useful: CVaR penalties and more elaborate scenario models do not automatically improve realised economics.

The modelling phase is sufficiently complete for the next work to focus on operability: packaging the decision application, scheduling it, monitoring data and model drift, comparing predicted and realised economics, and safely switching between champion models and robust fallbacks.

Supporting project documents

- [Frozen 2024-to-2025 forecast evaluation](#)
- [Final deterministic economic formulation](#)
- [Residual-bootstrap stochastic backtest](#)
- [Technical architecture](#)
- [Data-quality report](#)